

CELL-ME: Supervised Cell-Centered Graph Learning for Microenvironment Annotation and Downstream Spatial Analysis

Author Name

May 18, 2026

1 Introduction

Spatial omics technologies provide cell-resolved molecular measurements together with precise spatial coordinates, making it possible to study tissue organization directly *in situ* [1]. A central problem in this setting is microenvironment (ME) annotation: for each cell, the model must infer the local tissue context by combining information from the center cell with the composition and arrangement of its surrounding neighbors, where the center-cell information can be marker expression or cell-type identity depending on the feature setting. Reliable ME labels are important for identifying tumor regions, immune-enriched niches, stromal barriers, tertiary lymphoid structures, and tissue boundaries. However, manual ME annotation remains labor-intensive and difficult to scale across large cohorts or across slices with heterogeneous spatial patterns.

Graph-based modeling is a natural fit for this problem because each cell and its surrounding neighbors can be represented as a local spatial graph. Recent computational approaches have used graph deep learning, graph pooling, and spatial niche discovery to characterize tissue microenvironments or cellular neighborhoods from multiplexed spatial data [2, 3, 4]. Yet high-quality ME annotation requires more than learning generic graph representations. In practical use, the model must transfer across slices, remain robust to changing class balance, and preserve subtle neighborhood distinctions that separate related stromal and immune states. These requirements motivate a task-aligned framework that directly learns ME labels from local spatial graphs while maintaining sufficient representation stability for downstream retrieval, novelty detection, and biological interpretation.

To address this challenge, we introduce *CELL-ME*, an end-to-end supervised graph learning framework for cell-level microenvironment label prediction. CELL-ME represents each cell by a 2-hop neighborhood subgraph, augments node features with degree encoding, extracts local structural patterns with a Graph Isomorphism Network (GIN), summarizes each subgraph with attention pooling, and predicts the center-cell ME label using a lightweight multilayer perceptron (MLP). To improve robustness, CELL-ME additionally enforces consistency between original and augmented views of the same subgraph through a sign-regularization term (SigReg). The resulting model is trained directly for ME classification, while its learned embeddings remain useful for reference-query retrieval, withheld-class novelty analysis, and downstream biological analysis.

We evaluate CELL-ME on three spatial omics cohorts with curated ME annotations: a colorectal cancer (CRC) CODEX cohort, a triple-negative breast cancer (TNBC) CODEX cohort, and a hepatocellular carcinoma (HCC) spatial proteomics cohort derived from source studies [6, 7, 8]. Across these datasets, we first study standard held-out-slice ME annotation and embedding-based reference–query transfer. We then perform a CRC novelty analysis in which a follicle-associated microenvironment is withheld from the reference split. Finally, we ask a stricter biological question in CRC: although CELL-ME predictions under the limited-label setting are not identical to ground-truth ME labels, can the predicted labels still support the same types of downstream analyses that were originally performed with curated cellular neighborhoods? This final analysis tests whether predicted ME labels are useful not only as benchmark outputs, but also as interpretable units for spatial tissue biology.

Figure 1 summarizes the overall workflow. During training, labeled center-cell subgraphs from seen slices are used to optimize the encoder and classifier jointly. During inference, an unseen slice is converted to the same graph representation and passed through the trained model in a single forward pass, producing ME labels for all center cells. This design keeps the learning objective tightly matched to the final prediction task while preserving useful structural regularization from stochastic graph augmentation.

The main contributions of this work are threefold. First, we formulate ME annotation as direct supervised prediction on cell-centered neighborhood graphs, allowing the model to learn from local spatial context explicitly. Second, we introduce a compact graph architecture that combines degree encoding, GIN message passing, attention pooling, and SigReg-based augmentation consistency. Third, we show that CELL-ME supports not only accurate slice-level annotation and embedding-based retrieval across CRC, TNBC, and HCC, but also CRC-specific downstream analyses that recover known biological patterns despite prediction noise.

2 Method

2.1 Problem Setup and Graph Construction

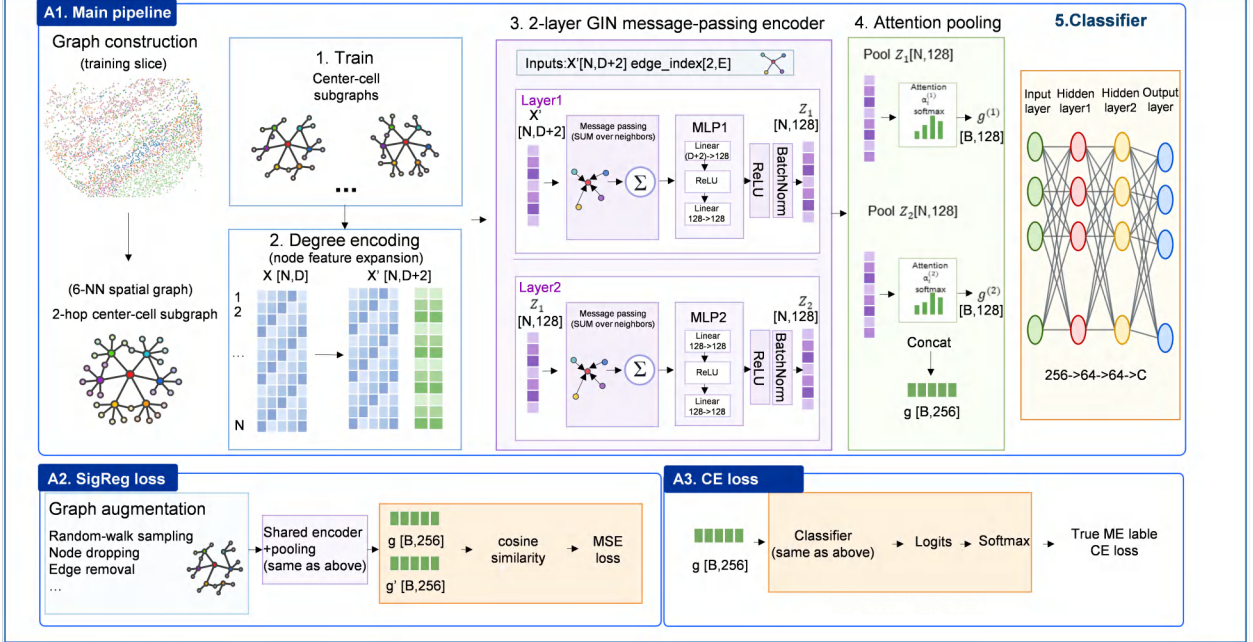
Let $\mathcal{S} = \{S^{(k)}\}_{k=1}^N$ denote a collection of spatial slices. Each slice $S^{(k)}$ contains n_k cells with spatial coordinates $p_i^{(k)} \in \mathbb{R}^2$, node features $x_i^{(k)} \in \mathbb{R}^D$, and, for training slices, a center-cell ME label $y_i^{(k)} \in \{1, \dots, C\}$. Depending on the feature setting, the node feature can be either a marker panel vector or a one-hot cell-type representation. In the experiments reported here, the downstream benchmarking and CRC analyses are conducted in the cell-type setting, which we denote as **ct** in the result plots.

For each slice we first build a slice-level spatial graph using a 6-nearest-neighbor rule in coordinate space. Formally, the k -th slice is written as

$$G_{\text{slice}}^{(k)} = (V^{(k)}, E^{(k)}),$$

where each node corresponds to a cell and an edge $(u, v) \in E^{(k)}$ indicates that cells u and v are spatial neighbors. Based on this slice graph, we extract a 2-hop induced subgraph for every center

A. Training graph construction and CELL-ME model



B. Application on unseen slice

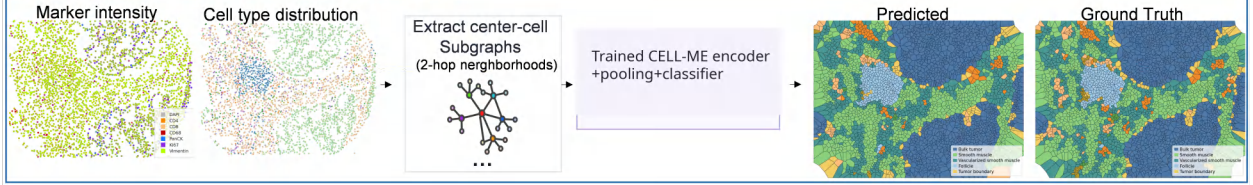


Figure 1: Overview of CELL-ME on CRC. Each center cell is represented by a 2-hop cell-neighbor subgraph extracted from a 6-nearest-neighbor spatial graph. The model appends structural degree features, encodes the original and augmented subgraphs with a shared GIN encoder, aggregates node information through attention pooling, predicts the center-cell ME label with an MLP classifier, and regularizes the two views with SigReg. The same encoder-classifier pipeline is then applied directly to unseen slices.

cell. For the i -th center cell, the resulting sample is

$$G_i = (V_i, E_i, X_i),$$

where V_i contains the center cell together with all cells reachable within two hops, E_i is the induced edge set, and $X_i \in \mathbb{R}^{|V_i| \times D}$ stacks the node features of all cells in the subgraph. Each graph sample inherits the ME label of its center cell.

This formulation offers two practical advantages. First, it preserves the local tissue context that is most relevant to ME recognition, including both cell composition and spatial arrangement. Second, it converts a large slice into a collection of aligned graph samples, allowing one shared model to be trained across all cells and slices.

2.2 CELL-ME Architecture

The overall architecture of CELL-ME is shown in Figure 1. Given a cell-neighborhood subgraph G_i , we first enrich each node feature with an explicit degree descriptor:

$$\tilde{x}_j = [x_j \parallel \log(1 + d_j^{\text{in}}) \parallel \log(1 + d_j^{\text{out}})],$$

where d_j^{in} and d_j^{out} denote the in-degree and out-degree of node j in the subgraph after adding self-loops for the encoding step, and $\parallel$ denotes concatenation. This degree encoding gives the encoder lightweight structural information that is not present in the raw panel or cell-type features alone.

The degree-augmented node features are then processed by a two-layer GIN encoder [5]. Let $H_i^{(0)} = \tilde{X}_i$. At the ℓ -th layer, the node embeddings are updated by

$$H_i^{(\ell)} = \text{GIN}^{(\ell)}(H_i^{(\ell-1)}, E_i), \quad \ell \in \{1, 2\}.$$

Each GIN layer uses 128 hidden channels. We obtain one graph summary from each layer and concatenate the two layerwise summaries to obtain a 256-dimensional graph-level representation. Compared with a plain neighborhood averaging scheme, this layered formulation preserves both lower-level and higher-level context within the same subgraph.

Instead of uniform mean pooling, CELL-ME uses attention pooling at each GIN layer. For layer ℓ with node embeddings $h_j^{(\ell)}$, we compute

$$\alpha_j^{(\ell)} = \frac{\exp(a_\ell(h_j^{(\ell)}))}{\sum_{u \in V_i} \exp(a_\ell(h_u^{(\ell)}))}, \quad g_i^{(\ell)} = \sum_{j \in V_i} \alpha_j^{(\ell)} h_j^{(\ell)},$$

and then form the final graph embedding as $g_i = [g_i^{(1)} \parallel g_i^{(2)}] \in \mathbb{R}^{256}$. Here $a_\ell(\cdot)$ is a learnable scoring network for layer ℓ . This layerwise attention pooling lets the model focus more strongly on informative neighbors, such as boundary cells or immune-enriched regions, instead of assigning identical weight to every node.

The graph embedding is passed to a supervised MLP classifier with dimensions $256 \rightarrow 64 \rightarrow 64 \rightarrow C$ to predict the ME label of the center cell:

$$\hat{y}_i = \text{MLP}(g_i).$$

This architecture directly optimizes the end task of center-cell ME-label prediction and avoids the extra representation-transfer step required by the original contrastive formulation. For each cohort, CELL-ME is trained as a closed-set classifier over that cohort’s own ME-label vocabulary, so a model trained on CRC predicts only CRC ME labels, while TNBC and HCC models use their own cohort-specific label spaces.

2.3 Training Objective and Inference

CELL-ME is trained end-to-end on labeled center-cell subgraphs from seen slices. During training, one branch is used to compute the consistency regularization on an original graph and a stochastically perturbed view of that same graph, while the supervised cross-entropy term is computed on

a second labeled mini-batch drawn from the same training split. In the current implementation, the perturbation branch uses identity plus a random choice among random-walk sampling, node dropping, and edge removal. The clean and perturbed graphs are encoded by the same network with shared parameters, producing graph embeddings for the two branches.

The primary optimization target is the standard cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{i=1}^B \log p(y_i | g_i),$$

where B is the batch size, y_i is the ground-truth ME label, and $p(y_i | g_i)$ is the predicted class probability from the MLP classifier. In implementation terms, this term is evaluated on the labeled classification mini-batch rather than on the same graph identities used by the consistency branch.

To stabilize the representation against local graph perturbations, we additionally apply sign regularization to the projected embeddings from the original and augmented views. Let $q_i, q'_i \in \mathbb{R}^d$ denote the projected embeddings of the two views of the same subgraph. The SigReg loss is defined as

$$\mathcal{L}_{\text{SigReg}} = \frac{1}{Bd} \sum_{i=1}^B \sum_{r=1}^d (\text{sign}(q_{ir}) - \text{sign}(q'_{ir}))^2.$$

This term encourages the embedding to preserve its coarse directional structure under modest perturbations, which improves robustness without introducing explicit positive-negative pair mining.

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{sig}} \mathcal{L}_{\text{SigReg}},$$

where λ_{sig} controls the strength of the regularization term. In the current implementation, λ_{sig} is fixed to 1. Unlike contrastive representation-learning objectives, CELL-ME uses direct supervised classification as the primary signal and keeps augmentation only as a lightweight consistency constraint.

At inference time, each unseen slice is converted to the same 6-nearest-neighbor graph and decomposed into 2-hop center-cell subgraphs. Every subgraph is then forwarded once through the trained encoder and classifier, and the predicted class is assigned as the ME label of the corresponding center cell. Because inference does not require co-training with the test slice, the resulting pipeline is simple, efficient, and naturally suited to slice-level generalization.

3 Model Evaluation and Representation Analysis

3.1 Datasets, Feature Setting, and Evaluation Protocol

We evaluate CELL-ME on three cohorts with curated ME labels: CRC, TNBC, and HCC. Unless otherwise specified, the learned graph model is reported in the cell-type feature setting (`ct`), where each node is represented by a one-hot cell-type vector and the downstream task is to predict the ME label of the center cell. For the standard annotation benchmark, slices are partitioned into

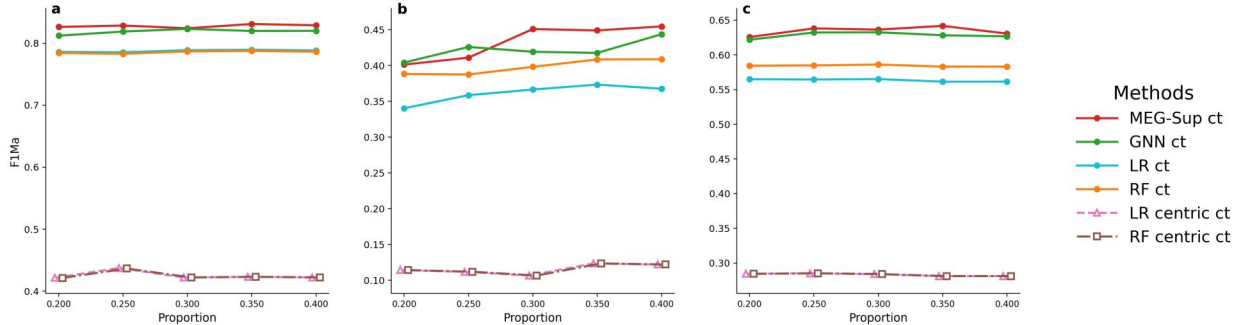


Figure 2: Cross-cohort benchmark on held-out slices. Panels (a)–(c) show macro-F1 performance on CRC, TNBC, and HCC, respectively, under the cell-type feature setting. Each cohort is trained and evaluated within its own label vocabulary under the same slice-wise split protocol, and higher values indicate better ME-label prediction.

seen slices used for optimization and unseen slices used only for evaluation. Only training slices are used for gradient-based optimization, validation slices are used for model selection, and test slices are held out for final evaluation. The graph encoder, attention pooling module, and MLP classifier are then applied directly to each unseen slice without test-time fine-tuning.

This same seen–unseen split also defines the retrieval-oriented analyses. After training, we encode all labeled center-cell subgraphs from the seen slices to build a reference bank in the learned embedding space. Cells from unseen slices are treated as query samples: each query subgraph is embedded once by the frozen model and then compared against the reference bank by cosine similarity. In the retrieval comparison, we additionally include two graph-level baselines: the mean one-hot cell-type composition within each neighborhood graph and the mean marker/panel expression within each neighborhood graph. This setup allows us to evaluate not only direct classification performance, but also whether the embedding space itself groups together neighborhoods with the same biological ME identity.

3.2 Cross-Cohort Benchmark on Held-Out Slices

Figure 2 summarizes the main benchmark across the three cohorts. The key observation is that separately trained CELL-ME models remain competitive across CRC, TNBC, and HCC while using a single shared graph construction and a single supervised training recipe. This consistency matters because the three cohorts differ substantially in tissue architecture, class balance, marker panel design, and local spatial complexity. In heterogeneous tissues such as CRC and TNBC, neighborhood context is especially useful for distinguishing morphologically adjacent but biologically different states, such as tumor-border, stromal, and immune-rich microenvironments. The benchmark is therefore consistent with the idea that direct supervision on cell-centered neighborhood graphs offers a practical tradeoff between expressive local context modeling and robust transfer to unseen slices.

Another useful takeaway from Figure 2 is that a method can obtain a favorable score on one cohort by exploiting dataset-specific composition biases, yet fail when tissue organization changes. In

contrast, the benchmark suggests that CELL-ME captures a reusable notion of local microenvironment structure across tissues. Even when the gap to competing methods is modest for a particular cohort, this across-cohort stability is valuable because it makes the resulting embedding space more trustworthy for downstream retrieval and novelty analyses.

3.3 Embedding-Based Transfer, Retrieval, and Novelty Analysis

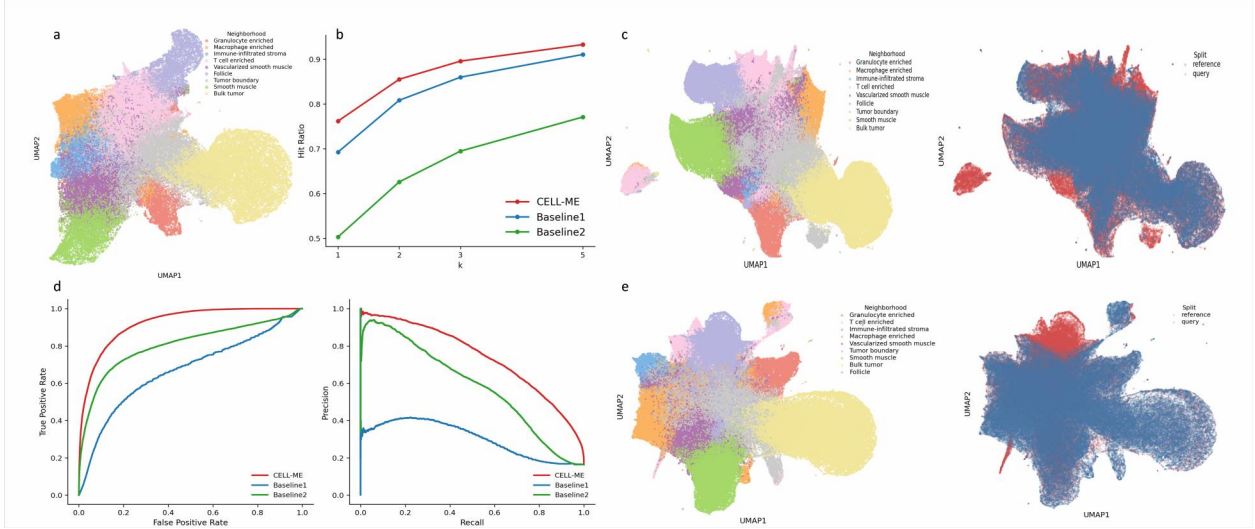


Figure 3: Embedding analysis and novelty detection on CRC. (a) UMAP of the CELL-ME reference embedding under the standard split, colored by center-cell ME label. (b) Reference-query retrieval performance measured by Hit Ratio@ K , comparing CELL-ME with a cell-type composition baseline (Baseline1) and a marker-expression baseline (Baseline2). (c) Joint UMAP of reference and query cells under the standard split, shown by ME label and by reference/query membership. (d) Follicle-held-out novelty detection using ROC and precision-recall curves, where Follicle query cells are positives and non-Follicle query cells are negatives. (e) UMAP of the Follicle-held-out novelty split, colored by ME label and by reference/query membership.

To test whether the learned embedding is useful beyond the classifier head, we perform explicit reference-query retrieval at the level of individual cell-centered neighborhood graphs (Figure 3(a)–(c)). The unit of retrieval is not an entire slice or tissue region, but a single graph sample built by taking a center cell, extracting its 2-hop spatial neighborhood, and assigning the sample the ME label of that center cell. Under the prop= 0.20 standard split, all train and validation graph samples are pooled into the reference bank, whereas all test graph samples form the query set. This yields 93,939 reference graphs and 159,880 query graphs. After row-wise ℓ_2 normalization, cosine similarity is computed between query and reference representations, and the top- K most similar reference graphs are retrieved for $K \in \{1, 2, 3, 5\}$.

Hit Ratio@ K is defined as a binary success indicator per query graph: if at least one of the top- K retrieved reference graphs has the same ME label as the query graph, the query counts as a hit. This metric therefore asks a concrete annotation-transfer question: does the retrieved reference neighborhood set contain at least one biologically matching exemplar? For a fair comparison,

the same retrieval pipeline is applied to three graph-level representations. CELL-ME uses the learned 256-dimensional graph embedding. The cell-type baseline averages the one-hot cell-type features over all nodes in each graph, which is equivalent to a vector of local cell-type proportions. The expression baseline averages marker/panel features over all nodes in each graph, producing a vector of local mean panel values. All three representations are then evaluated with the same reference/query split, the same row-wise normalization, the same cosine top- K retrieval rule, and the same Hit Ratio@ K metric.

Figure 3(b) shows that the learned CELL-ME representation performs best throughout the retrieval range. Hit Ratio@{1,2,3,5} is 0.7618, 0.8550, 0.8956, and 0.9325 for CELL-ME, compared with 0.6925, 0.8082, 0.8597, and 0.9105 for the cell-type baseline and 0.5030, 0.6259, 0.6946, and 0.7707 for the expression baseline. The advantage is most pronounced at small K , indicating that the learned embedding organizes local microenvironments more sharply than simple neighborhood composition or mean-expression summaries. The UMAP visualizations support the same conclusion: the reference-side embedding already separates most ME classes, and the joint reference–query embedding preserves this organization after held-out slices are added (Figure 3(a),(c)).

We further ask whether the embedding can flag a microenvironment that is absent from the reference ontology (Figure 3(d),(e)). For this analysis, all CRC slices with zero Follicle cells are assigned to the reference side, whereas all slices containing at least one Follicle cell are assigned to the query side. The zero-Follicle slices are further divided into train, validation, and reference-extra subsets. Only the train subset is used for gradient-based optimization, the validation subset is used for model selection, and the final reference bank is constructed after training from embeddings of all cells in the train, validation, and reference-extra zero-Follicle slices. The query set contains all cells from Follicle-positive slices, so it includes both withheld Follicle cells and known non-Follicle microenvironments from the same tissue sections. This design removes Follicle not only from the reference bank but also from supervised model fitting, making novelty detection a strict held-out-class test rather than an artificial outlier exercise.

After training with zero-Follicle slices and selecting the checkpoint with the validation split, we discard the classifier head for novelty scoring and use the frozen encoder to extract embeddings for all reference-side and query-side cells. Let $R \in \mathbb{R}^{n_r \times d}$ denote the matrix of embeddings from all zero-Follicle reference cells and $Q \in \mathbb{R}^{n_q \times d}$ the matrix of embeddings from all cells in Follicle-positive query slices. Each row is normalized to unit Euclidean norm, yielding $\bar{R}$ and $\bar{Q}$. For the m -th query cell, the implemented novelty score is

$$s_{\text{novel}}(\bar{q}_m) = 1 - \max_{1 \leq j \leq n_r} \bar{q}_m^\top \bar{r}_j,$$

where $\bar{q}_m$ is the normalized embedding of the m -th query cell and $\bar{r}_j$ is the normalized embedding of the j -th reference cell. Because both rows are ℓ_2 normalized, the inner product is exactly cosine similarity, so the score is precisely one minus the best reference cosine match. No top- k averaging, temperature scaling, softmax conversion, or additional z -score normalization is applied.

Positive samples are query cells whose ground-truth ME label is Follicle. Negative samples are all remaining non-Follicle query cells from the same Follicle-positive slices. ROC and precision–recall curves are then obtained by sweeping a threshold directly over the cell-level scores s_{novel} , without any slice-level, region-level, or cluster-level aggregation. Under this setup, CELL-ME achieves a ROC-AUC of 0.9232 and a PR-AUC of 0.7422, compared with 0.6671 and 0.3013 for the cell-type

baseline and 0.8049 and 0.5849 for the expression baseline. This setup is more informative than simply checking whether the classifier mislabels withheld cells, because it evaluates whether the embedding space learned from known microenvironments preserves a usable notion of *distance from known biology*. The resulting separation suggests that CELL-ME learns a representation capable of both assigning known labels and flagging coherent unseen microenvironments when the reference ontology is incomplete.

4 CRC Spatial-Ecology Readouts from Predicted ME Labels

4.1 Evaluation Rationale and CRC Reference Framework

The analyses above evaluate CELL-ME as a prediction and representation-learning model. We next ask whether its predicted CRC ME labels are biologically useful as analysis units. This question is important because the model is not perfect: under the CRC prop= 0.20 setting used here, the predicted labels cover 253,819 cells from 35 patients and reach a cell-level accuracy of 0.8222 overall, with 0.8374 in CLR patients and 0.8108 in DII patients. Thus, the predicted labels contain nontrivial noise and are not a one-to-one replacement for curated ground truth. The relevant downstream question is therefore whether a noisy predicted label field can still preserve the major spatial-ecological signals that are needed for cohort-level biological interpretation.

To define this downstream evaluation in a biologically grounded way, we adapt the spatial-ecology framework established in the CRC CODEX study by Schürch et al. [6]. That work used cellular neighborhoods (CNs) to connect local cell-type composition, patient-group structure, tissue modules, T-cell functional states, and cross-neighborhood associations in the colorectal cancer invasive tumor microenvironment. Here, we use CELL-ME predicted MEs as the corresponding spatial units and examine whether they support the same classes of readouts. This is not intended as an exact re-annotation of the original CN pipeline; rather, it asks whether supervised graph-predicted MEs retain enough biological structure to act as interpretable spatial units when manual or curated ME labels are only partially available.

4.2 Immune Composition and Patient-Level ME Abundance

We first examine whether the predicted labels recover the known immunological contrast between Crohn’s-like lymphoid reaction (CLR) and diffuse inflammatory infiltration (DII). At the immune-cell composition level, CLR tumors show a pronounced enrichment of B cells, whereas DII tumors contain higher fractions of macrophages and granulocytes (Figure 4(a)). This establishes the expected biological background: CLR is associated with organized lymphoid structure, while DII is associated with a more diffuse myeloid- and granulocyte-rich inflammatory state. Although this panel does not use predicted ME labels directly, it provides the cellular context in which the subsequent predicted-ME analyses should be interpreted.

CELL-ME predicted ME frequencies organize this immune-cell contrast into spatial microenvironment labels (Figure 4(b),(c)).

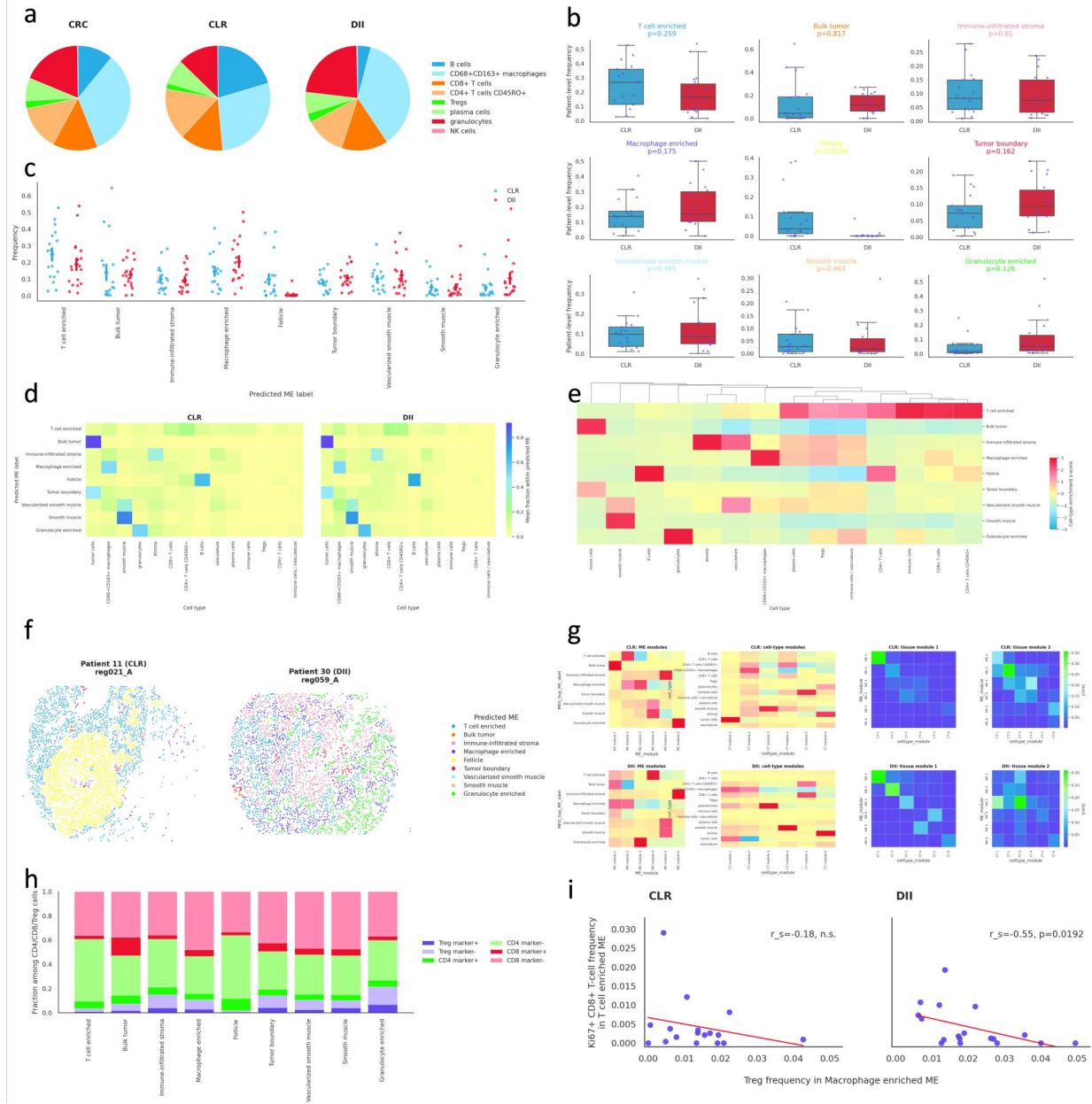


Figure 4: CRC spatial-ecology readouts using CELL-ME predicted ME labels from the prop= 0.20 setting. (a) Immune-cell composition in all CRC samples, CLR tumors, and DII tumors. (b) Patient-level frequencies of each predicted ME label in CLR versus DII, with group-comparison p values shown in each panel. (c) Patient-level dot plot of predicted ME frequencies across all ME labels, highlighting between-patient heterogeneity. (d) Cell-type composition within each predicted ME label, shown separately for CLR and DII. (e) Cell-type enrichment z-scores across predicted ME labels. (f) Representative spatial maps of predicted ME labels in a CLR slice and a DII slice. (g) HOSVD/Tucker-style modules from patient $\times$ predicted-ME $\times$ cell-type tensors in CLR and DII. (h) T-cell functional-state composition within predicted ME labels. (i) ME-specific correlation between Treg frequency in macrophage-enriched MEs and Ki67+ CD8+ T-cell frequency in T-cell-enriched MEs, shown separately for CLR and DII.

The strongest patient-group signal is the Follicle ME, whose mean frequency is substantially higher in CLR than in DII (9.70% versus 0.62%, FDR= 0.0138). This direction matches the central CRC observation that CLR tumors contain organized lymphoid follicle-like structures, whereas DII tumors largely lack them. In contrast, DII tumors show non-significant but biologically consistent trends toward higher macrophage-enriched, granulocyte-enriched, and tumor-boundary MEs. The patient-level dot plot adds an important distributional view: Follicle depletion in DII is observed across the cohort rather than being driven by one or two outlier patients, while the myeloid- and granulocyte-associated MEs show broader inter-patient heterogeneity.

4.3 Predicted-Label Semantics and Spatial Coherence

The next question is whether the predicted labels have stable semantic content. The cell-type composition heatmaps show that the major predicted labels are dominated by their expected lineages in both CLR and DII (Figure 4(d)). Bulk tumor is almost entirely composed of tumor cells, Follicle is B-cell rich, Macrophage enriched is dominated by CD68+CD163+ macrophages, Granulocyte enriched is dominated by granulocytes, and Smooth muscle is dominated by smooth muscle cells. This result is important because the downstream analyses would be difficult to interpret if the predicted ME names were disconnected from their cellular composition.

The enrichment heatmap provides a complementary view of the same semantic stability (Figure 4(e)). Whereas the composition heatmap asks which cell types numerically dominate each ME, the enrichment view asks which lineages are selectively overrepresented relative to the overall background. These two views are not redundant. For example, tumor cells may be present in several tissue contexts, but the Bulk tumor label is distinguished by strong tumor-cell enrichment, while Tumor boundary represents a more mixed interface state. The agreement between composition and enrichment therefore supports the use of CELL-ME predicted MEs as interpretable microenvironment categories rather than arbitrary classifier outputs.

Spatial coherence provides a further validation layer. Representative maps show that a CLR slice is dominated by contiguous Follicle and T cell enriched regions, consistent with organized lymphoid architecture (Figure 4(f)). In contrast, a representative DII slice lacks a dominant Follicle structure and instead contains a diffuse mixture of macrophage-enriched, immune-infiltrated stromal, T-cell-enriched, granulocyte-enriched, and tumor-boundary MEs. This distinction is stronger than a simple immune-versus-nonimmune comparison: the predicted maps suggest that CELL-ME captures differences in spatial organization, not only differences in overall immune-cell abundance.

4.4 Higher-Order Tissue Modules and Functional Associations

We then move from single-label summaries to higher-order tissue organization. Motivated by the tensor-decomposition analysis in the CRC CODEX study [6], we construct patient $\times$ predicted-ME $\times$ cell-type tensors and apply an HOSVD/Tucker-style decomposition separately in CLR and DII (Figure 4(g)). In CLR, the resulting modules separate tumor, immune/myeloid-stromal, and structural compartments. In DII, the dominant axes emphasize macrophage, tumor, and granulocyte coupling. Because this is an exploratory decomposition in a 35-patient cohort, individual module

signs should not be overinterpreted. Nevertheless, the result shows that predicted MEs can support multiway tissue-module analysis, which is a more stringent requirement than recovering marginal ME frequencies alone.

Finally, predicted MEs retain enough structure to support functional-state and cross-ME association analyses (Figure 4(h),(i)). T-cell functional-state composition varies across predicted MEs: granulocyte-enriched and tumor-boundary MEs contain higher fractions of Treg-associated states, whereas Follicle MEs are dominated by CD4/CD8 states with comparatively low Treg marker-positive fractions. A ME-specific correlation analysis further reveals a DII-specific negative association between Treg frequency in macrophage-enriched MEs and Ki67+ CD8+ T-cell frequency in T-cell-enriched MEs (Spearman $r_s = -0.545$, $p = 0.0192$), whereas no such association is observed in CLR tumors. This does not establish causality, but it parallels the type of group-specific functional coordination analyzed in the original CRC CN framework: in DII, macrophage/Treg-associated niches are linked at the patient level to reduced proliferative CD8+ T-cell state in T-cell-enriched regions.

Taken together, Figure 4 evaluates predicted ME labels across a progression of increasingly demanding biological readouts: immune composition, patient-level ME abundance, label semantics, spatial coherence, multiway tissue modules, functional-state composition, and cross-ME correlation. The consistent recovery of CLR-associated Follicle/T-cell organization and DII-associated macrophage/granulocyte inflammatory structure indicates that CELL-ME predictions remain biologically meaningful even when cell-level prediction is imperfect. Thus, the predicted ME field is not only a target for macro-F1 benchmarking; it can also serve as a practical substrate for downstream spatial-ecology analysis in settings where complete manual ME annotation is unavailable.

5 Conclusion and Discussion

This study presents CELL-ME as a task-aligned graph-learning framework for microenvironment annotation in spatial omics. By converting each center cell into a local 2-hop neighborhood graph and optimizing the encoder directly for supervised ME prediction, CELL-ME ties representation learning tightly to the final biological objective. The experiments show that this design is useful at three levels: it supports accurate held-out-slice annotation across CRC, TNBC, and HCC; it organizes a reference-query embedding space that remains informative for nearest-neighbor retrieval and novelty detection; and it produces predicted CRC ME labels that remain meaningful for downstream biological analysis despite imperfect cell-level prediction.

The broader implication is that local graph supervision can serve as more than a predictive shortcut. When the embedding is stable under modest graph perturbations and remains interpretable through retrieval, it becomes a practical substrate for downstream analysis rather than a disposable intermediate feature. That property is especially relevant in spatial biology, where analysts often need to move fluidly between automated labels, neighborhood exemplars, and exploratory inspection of unusual tissue regions.

At the same time, several limitations remain. The current study uses a fixed 6-nearest-neighbor graph and a fixed 2-hop receptive field, which may not capture the most informative spatial scale

for every tissue or assay. The main results reported here also focus on the cell-type feature setting, leaving multimodal marker-rich inputs for future comparison. A natural next step is therefore to extend CELL-ME toward adaptive neighborhood construction, richer uncertainty calibration, and cross-platform transfer across assays with partially mismatched feature spaces. Even in its current form, however, CELL-ME provides a compact and effective framework for supervised ME annotation together with retrieval-, novelty-, and downstream-analysis-oriented use cases.

References

- [1] Sabrina M. Lewis, Marie-Liesse Asselin-Labat, Quan Nguyen, Jean Berthelet, Xiao Tan, Verena C. Wimmer, Delphine Merino, Kelly L. Rogers, and Shalin H. Naik. Spatial omics and multiplexed imaging to explore cancer biology. *Nature Methods*, 18:997–1012, 2021. doi:10.1038/s41592-021-01203-6.
- [2] Zhenqin Wu, Alexandro E. Trevino, Eric Wu, Kyle Swanson, Honesty J. Kim, H. Blaize D’Angio, Ryan Preska, Gregory W. Charville, Piero D. Dalerba, Ann Marie Egloff, Ravindra Uppaluri, Umamaheswar Duvvuri, Aaron T. Mayer, and James Zou. Graph deep learning for the characterization of tumour microenvironments from spatial protein profiles in tissue specimens. *Nature Biomedical Engineering*, 6:1435–1448, 2022. doi:10.1038/s41551-022-00951-w.
- [3] Yuxuan Hu, Jiazhen Rong, Runzhi Xie, Yafei Xu, Jacqueline Peng, Lin Gao, and Kai Tan. Learning predictive models of tissue cellular neighborhoods from cell phenotypes with graph pooling. *bioRxiv*, 2022. doi:10.1101/2022.11.06.515344.
- [4] Marco Varrone, Daniele Tavernari, Albert Santamaria-Martínez, Logan A. Walsh, and Giovanni Ciriello. CellCharter reveals spatial cell niches associated with tissue remodeling and cell plasticity. *Nature Genetics*, 56:74–84, 2024. doi:10.1038/s41588-023-01588-4.
- [5] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations*, 2019.
- [6] Christian M. Schürch, Salil S. Bhate, Graham L. Barlow, Darci J. Phillips, Luca Noti, Inti Zlobec, Pauline Chu, Sarah Black, Janos Demeter, David R. McIlwain, Shigemi Kinoshita, Nikolay Samusik, Yury Goltsev, and Garry P. Nolan. Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front. *Cell*, 182(5):1341–1359.e19, 2020. doi:10.1016/j.cell.2020.07.005.
- [7] Stephen L. Shiao, Kenneth H. Gouin III, Nathan Ing, et al. Single-cell and spatial profiling identify three response trajectories to pembrolizumab and radiation therapy in triple negative breast cancer. *Cancer Cell*, 42:70–84, 2024. doi:10.1016/j.ccell.2023.12.012.
- [8] Zhenqin Wu, Joseph Boen, Sonali Jindal, Sreyashi Basu, Matthew Bieniosek, Siyu He, Michael LaPelusa, Aaron T. Mayer, Ahmed O. Kaseb, James Zou, Padmanee Sharma, and Alexandro E. Trevino. Spatial multi-omics and deep learning reveal fingerprints of immunotherapy response and resistance in hepatocellular carcinoma. *bioRxiv*, 2025. doi:10.1101/2025.06.11.656869.